芯语智分

Xinyu SmartSort

A Voice-Visual Waste Sorting System
Based on Zynq Hardware-Software Co-Design

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Z19 / Zynq UltraScale+ MPSoC

AMD

01 WHY IT MATTERS

Hear. See. Sort.

Turn waste sorting into a visible experience. A voice command starts recognition, spoken feedback and a robotic sorting action.

ONE COMMAND, ONE TASK

1. **Wake Xiaoya**
   "你好小亚"
2. **Start sorting**
   "开始垃圾分拣"
3. **Prepare and capture**
   Camera view
4. **Recognize the category**
   Visual detection
5. **Announce the result**
   Spoken feedback
6. **Pick and place**
   Fixed pickup station
7. **Return to waiting**
   Ready for the next task

02 ZYNQ CO-DESIGN

Four-function source architecture · functional roles

PS | Visual intelligence
Camera capture · YOLO on CPU
Category + coordinate recovery

PS | Speech intelligence
USB audio capture · Vosk ASR
Wake and task command events

PL | Video + candidate rules
RGB / HSV · ROI statistics
First moments · candidate checks

PL | Audio frontend
DC suppression · 63-tap FIR
Filtering before speech recognition

PS TASK SERVICE | Coordinates · categories · commands

VM / HOST | Task, voice and motion
Task management · announcement · SDK actions · visualization

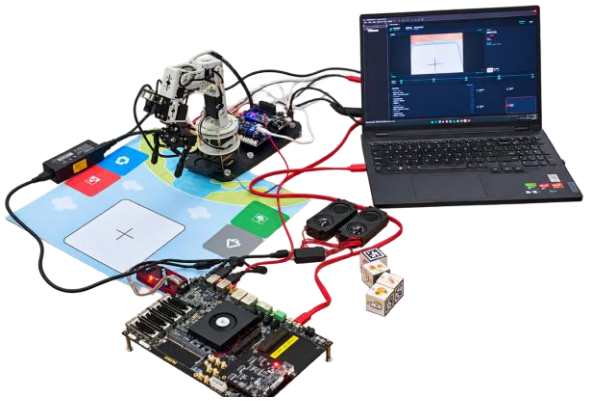
DESIGNED TO WORK TOGETHER

Natural interaction
One wake-and-start cycle links recognition, feedback and action.

Clear hardware-software roles
PS runs models and services; PL processes pixel and audio streams.

Modular engineering
Four functions support reuse, teaching and further integration.

03 FROM IDEA TO ACTION



DOFBOT · sorting station · Z19 · laptop host

PLATFORM AT A GLANCE

Core	ALINX Z19 · Zynq UltraScale+ MPSoC
Compute	PS + PL + Ubuntu VM
Devices	Camera · voice input · DOFBOT
Stack	Linux · Python · YOLO · Vosk

DEMONSTRATED & DELIVERED

Fixed-station VM / DOFBOT demonstration.
Z19 PS/PL source modules with clear interfaces.
Next: joint board integration and wider object coverage.

Make sorting visible. Make engineering tangible.

Campus sustainability / Community outreach / Embedded education